Exercises · Chapter 2: Probability

Exercises with answer options (a)–(d) are **single-choice** questions: exactly one answer is correct.

1. **Two Coin Tosses** (Script: Exercise "Two Consecutive Coin Tosses")

   A coin lands heads with probability $P(H) = 0.7$. The coin is tossed twice, and the two tosses are independent. Let H_1 be the event that heads occurs on toss 1 and let H_2 be the event that heads occurs on toss 2.

   What is the probability that heads occurs on toss 1 **or** on toss 2, i.e., $P(H_1 \cup H_2)$?

   (a) 0.49
   (b) 0.42
   (c) 0.51
   (d) 0.91

2. **Disjoint Events** (Script: Exercise "Independence vs. Disjoint Events")

   A shop's customers pay by card (event A), in cash (event B), or with their mobile phone (event C). Each customer uses exactly one payment method, such that the three events are disjoint: $A \cap B = \emptyset$, $A \cap C = \emptyset$, and $B \cap C = \emptyset$. Moreover,

   $$P(A) = 0.4 \quad \text{and} \quad P(B) = 0.5.$$

   Which of the following statements is true?

   (a) $P(A \cup B) = P(A)\,P(B) = 0.20$.
   (b) A and B are independent, since $P(A \cap B) = P(A)\,P(B) = 0.20$.
   (c) A and B are *not* independent, since $P(A \cap B) = 0 \neq 0.20 = P(A)\,P(B)$.
   (d) $P(A \mid B) = P(A) = 0.4$.

3. **Independent Events?** (Script: Exercise "Deriving Independence")

   A fair die is tossed once. Let

   $$A = \{2, 4, 6\}$$

   be the event of observing an even number and let

   $$B = \{1, 2\}.$$

   Which of the following statements is true?

   (a) $P(A \cup B) = P(A) + P(B)$.
   (b) A and B are disjoint, i.e., $A \cap B = \emptyset$.
   (c) A and B are independent.
   (d) A and B are not independent.

4. **Medical Test** (Script: Example "Medical Tests")

   A disease D occurs in 1% of the population, i.e., $P(D) = 0.01$. A medical test for this disease has outcomes $+$ and $-$, with

   $$\text{sensitivity:} \quad P(+ \mid D) = 0.95,$$

   $$\text{specificity:} \quad P(- \mid D^c) = 0.9.$$

   A person gets a positive test result. What is the probability that this person has the disease, i.e., $P(D \mid +)$? (Answers are rounded to three decimals.)

(a) 0.088

(b) 0.900

(c) 0.109

(d) 0.950

5. **Drawing Cards** (Script: Exercise "Drawing Cards Without Replacement")

   Draw two cards from a standard deck of 52 cards *without* replacement. The deck contains 4 aces. What is the probability that **both** cards are aces, i.e., that the first card is an ace **and** the second card is an ace? (Answers are rounded to four decimals.)

   (a) $\frac{4}{52} \cdot \frac{3}{51} \approx 0.0045$

   (b) $\frac{4}{52} \cdot \frac{4}{51} \approx 0.006$

   (c) $\frac{3}{51} \approx 0.0588$

   (d) $\frac{4}{52} \cdot \frac{4}{52} \approx 0.0059$

6. **pmf, cdf, Mean, and Variance** (Script: Examples "cdf of a Discrete Random Variable (Car Sales)" and "pmf of a Discrete Random Variable (Car Sales)")

   Another car dealership sells $X \in \{0, 1, 2, 3\}$ cars per day with the following probability mass function (pmf):

x	0	1	2	3
$f_X(x) = P(X = x)$	0.4	0.3	0.2	0.1

   (a) Check that f_X is a valid pmf.

   (b) Write down the cdf $F_X(x) = P(X \leq x)$ for all $x \in \mathbb{R}$ and sketch it.

   (c) Compute the probability that the dealership sells at most one car on a given day, and the probability that it sells more than two cars.

   (d) Compute the mean $E(X)$, i.e., the average number of cars sold per day.

   (e) Compute the variance $\mathrm{Var}(X)$.

7. **Waiting Times** (Script: Examples "cdf of a Continuous Random Variable (Customer Arrivals)" and "pdf of a Continuous Random Variable (Customer Arrivals)")

   Let $X \in [0, \infty)$ be the waiting time (in minutes) until the next customer calls a company's service hotline. The waiting time is exponentially distributed with an average waiting time of 10 minutes, i.e., its cdf is

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - e^{-x/10} & \text{for } x \geq 0. \end{cases}$$

   (a) Compute the probability that the next customer calls within 5 minutes.

   (b) Compute the probability that the hotline has to wait more than 30 minutes for the next call.

   (c) Compute the probability that the waiting time is between 5 and 30 minutes.

   (d) Derive the density function f_X of X.

   (e) Show that the probability in (a) equals the area under the density f_X between 0 and 5.

8. **Joint pmf, Conditional Means, and Independence** (Script: Examples "Two-Dimensional pmf (Car Sales and Staffing)", "Law of Total Expectation (Car Sales and Staffing)", and "Checking Independence in Our Examples")

   Let $X_1 \in \{0, 1, 2, 3\}$ be the number of cars sold per day by a car dealership, and let $X_2 \in \{1, 2\}$ be the number of sales consultants on duty that day. The joint pmf $f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$ is given by the following table:

$f(x_1, x_2)$	$x_1 = 0$	$x_1 = 1$	$x_1 = 2$	$x_1 = 3$
$x_2 = 1$	0.20	0.20	0.10	0.00
$x_2 = 2$	0.05	0.15	0.20	0.10

(a) Compute the marginal pmf f_1 of X_1 and the marginal pmf f_2 of X_2.

(b) Compute the conditional means $E(X_1 \mid X_2 = 1)$ and $E(X_1 \mid X_2 = 2)$. Interpret the result.

(c) Compute the unconditional mean $E(X_1)$ in two ways: (i) using the marginal pmf f_1, and (ii) using the Law of Total Expectation.

(d) Are X_1 and X_2 independent? Justify your answer.

9. **Two-Dimensional Density** (Script: Example "Two-Dimensional pdf (Customer Arrivals and Service Times)")

   A company runs a service hotline. Let

   - $X_1 \geq 0$ be the waiting time (in minutes) until the next call, and
   - $X_2 \geq 0$ be the duration (in minutes) of this call.

   The random vector $X = (X_1, X_2)^\top$ has the joint density function

$$f(x_1, x_2) = \begin{cases} \frac{1}{6}\, e^{-x_1/6} \cdot \frac{1}{3}\, e^{-x_2/3} & \text{for } x_1 \geq 0 \text{ and } x_2 \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

   (a) Show that f integrates to one.

   (b) Compute the probability that the next call comes within 6 minutes **and** lasts at most 6 minutes.

   (c) Compute the marginal density $f_1(x_1)$ of the waiting time X_1.

   (d) Are X_1 and X_2 independent? Justify your answer.

10. **Regression Function** (Script: Example "Conditional Mean Function of a Bivariate Normal Distribution")

    The revenue growth X_1 of a firm and the GDP growth X_2 (both in %) are jointly normally distributed with

$$\mu = \begin{pmatrix} E(X_1) \\ E(X_2) \end{pmatrix} = \begin{pmatrix} 2 \\ 0.5 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_1, X_2) & \mathrm{Var}(X_2) \end{pmatrix} = \begin{pmatrix} 1 & 0.4 \\ 0.4 & 2 \end{pmatrix}.$$

    For the bivariate normal distribution, the regression function is

$$m(x_2) = E(X_1 \mid X_2 = x_2) = E(X_1) + \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_2)} \big(x_2 - E(X_2)\big).$$

    Which of the following is the regression function $m(x_2)$?

    (a) $m(x_2) = 1.90 + 0.20\, x_2$

    (b) $m(x_2) = 0.10 + 0.20\, x_2$

    (c) $m(x_2) = 1.80 + 0.40\, x_2$

    (d) $m(x_2) = 1.86 + 0.28\, x_2$

11. **Covariance, Regression Function, and the Law of Total Expectation** (Script: Examples "Covariance and Correlation of a Bivariate Normal Distribution", "Conditional Mean Function of a Bivariate Normal Distribution", and "Law of Total Expectation (Bivariate Normal Distribution)")

A car manufacturer's revenues react strongly to the business cycle. Let X_1 be the manufacturer's annual revenue growth and let X_2 be the annual GDP growth (both in %). Suppose that

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim \mathcal{N}_2\left(\mu, \Sigma\right) \quad \text{with} \quad \mu = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 4 & 1.2 \\ 1.2 & 1 \end{pmatrix}.$$

(a) What are the standard deviations of X_1 and X_2? Compute the correlation $\mathrm{Corr}(X_1, X_2)$.

(b) Compute the regression function $m(x_2) = E(X_1 \mid X_2 = x_2)$ using

$$m(x_2) = E(X_1) + \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_2)}\big(x_2 - E(X_2)\big).$$

(c) Interpret the slope of the regression function. What is the expected revenue growth in a year with zero GDP growth, and in a year with 3 percent GDP growth?

(d) Verify the Law of Total Expectation, $E\big(E(X_1 \mid X_2)\big) = E(X_1)$, for this example.

12. **Mean Independence vs. Independence** (Script: Section "Independence Implies Mean Independence (but Not Vice Versa)")

Let X_1 be the daily return of a stock (in %), and let $X_2 \in \{1, 2\}$ describe the state of the stock market on that day: calm ($X_2 = 1$) or volatile ($X_2 = 2$), where

$$P(X_2 = 1) = 0.8 \quad \text{and} \quad P(X_2 = 2) = 0.2.$$

Given the state of the market, the return is normally distributed:

$$X_1 \mid X_2 = 1 \sim \mathcal{N}\left(0.2,\, 1\right),$$

$$X_1 \mid X_2 = 2 \sim \mathcal{N}\left(0.2,\, 4\right).$$

(a) Compute the conditional means $E(X_1 \mid X_2 = 1)$ and $E(X_1 \mid X_2 = 2)$.

(b) Compute the unconditional mean $E(X_1)$ using the Law of Total Expectation.

(c) Is X_1 mean independent of X_2?

(d) Compute the conditional variances $\mathrm{Var}(X_1 \mid X_2 = 1)$ and $\mathrm{Var}(X_1 \mid X_2 = 2)$. Are X_1 and X_2 independent? Interpret the result.